

Detailed Connections

MicroSD Card Reader Pins to MKR 1310 Pins Connections

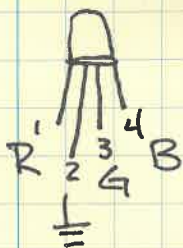
3V → VCC
 CLK → SCK (Pin#9)
 SO → MISO (Pin#10)
 SI → MOSI (Pin#8)
 CS (chipSelect) → Pin#11

1.5A Battery Charger Connections to MKR 1310 Pin Connections

CE (Charge Enable) → Pin#6
 LiPo → Analog Pin#6 Via Voltage divider that goes to ground.

- JST Male / Male Connector (top Side Female Connection Port)
Connects from JST 2-Pin Breakout Connector Port to 1.5A LiPo Charger Load Out JST Port
- JST Male Battery Connector (top Side Female connection Port)
Connects to 1.5A LiPo Charger LiPo Batt JST Port
- 6V Mini Solar Panel Connected to 1.5A LiPo Charger
Using 2.1mm Barrel Plug into 10V Max (topside) Barrel Plug Receiver.

Tri Color LED



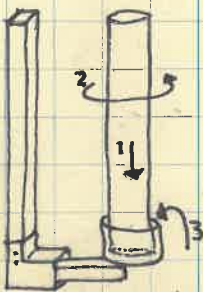
- Soldered in place with 3x 220Ω Resistors
- | Arduino Pin | LED |
|-------------|---------|
| #0 | → Red |
| #1 | → Green |
| #4 | → Blue |
- (Note: In the original image, pins #0, #1, and #4 are grouped together with a bracket and labeled 'Digital'.)*

See Page 53 For Colors Code.

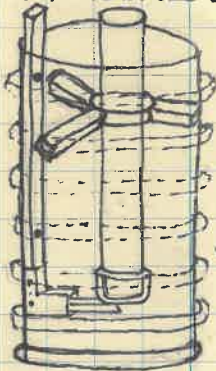
For Any adjustments to the Component Layout
of the Board

1. Remove On/Off Switch
2. Add Switch to Run Calibrate mode / Run mode
3. Add Led to Indicate Calibration mode (Field)
 - Tare Red
 - Weighted Tare Blue
 - Run mode Green

6/2/2025 Assembly Continued (Part by Part)

glue inside covering
edges and screwsSpread glue
to cover all the way
around end of PVC

1. Insert pipe into cap flush with bottom
2. Gently turn pipe to spread glue evenly
3. Smooth down and clean top edge of cap around pipe to make a good seal



Put Full Assembly of Aluminum Rod support Back into corrugated Pipe attach Rod and set upright for glue to dry

Attach PVC Pipe to Bottom Cap (This cannot be Undone)

- Remove Aluminum Rod with Loadcell assembly from Corrugated Pipe.
- Mix JB Weld Plastic Bonder and apply to Both the inside of the Bottom Cap (Cover screws on inside) and the outside of the PVC Pipe at the Bottom.
- Push the PVC Pipe into the cap and spingently to spread adhesive evenly.
- Smooth the adhesive around the pipe and Remove any excess.
- Make sure there are no bubbles and the adhesive is covering the entire junction
- Carefully Guide the Aluminum Rod Assembly and PVC pipe into the corrugated Pipe and Re attach the Aluminum Rod in place with the PVC Pipe guided through the Support.
- Set the Corrugated pipe Upright. Ensure the PVC Pipe remains flush with the Bottom of the Support Cap and allow the Glue/Adhesive to fully Cure/dry.

Attach Top Cap to PVC PipeThis is a Point of No Return

Once this is done the full assembly of the Aluminum Rod/ Pipe Support/ PVC Pipe Load cell and Corrugated pipe is Permanently Together and cannot Be taken Apart

The Top Cap is a Manufactured PVC Cap. It is glued in place using...

- Oatey Purple Primer
- Oatey Regular PVC Cement

Follow the Manufacturers Directions for Use.

Screw Eyes
Armellas

Zinc / Cinc

#212 x 15/16 IN

9 PCS
PZS

490967

HILLMAN



Assembly Should Look Like above before attaching top Cap.

- Top Cap Should have 15/16" Hole drilled in Center to attach Screw eye to. Caulk inside around Screw B-4 attaching.

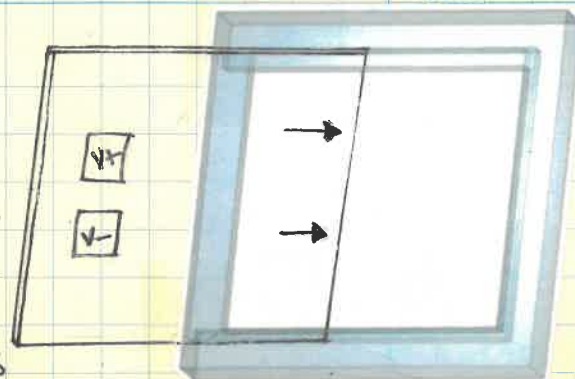
6/3/2025

Assembly Continued (Part by Part)

Solar Panel Assembly

6V 0.6W Mini Solar Panel

- Slide Solar Panel into 3-D Printed Holder with Solderpads facing Bottom of Holder (3-Sided)



- Solder the Waterproof Cable ends to the V+, V- Pads on the Back side of the panel. (DC Power Cable Set Waterproof Adafruit Part # 743)

★ This wire set is too Short and must be extended to allow the Components Box to still open Once the Solar Panel is permanently mounted to the top of the Box.



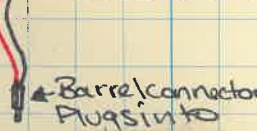
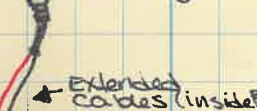
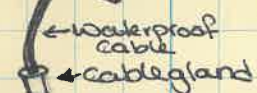
← This Side Outside Box Must Be Toward Solar Panel

← Has to be able to fit over Barrel Connector

- Clip the Male 2.1mm Barrel Connector off with 1/2" of cable attached Strip the wires on Both Sides.

→ Trim down the length of the Barrel connector and reduce its ~~width~~ circumference to allow it to slip through the Cable gland Lock nut.

- Slide the M12 cable gland on to the waterproof cable with the tightener toward the Solar panel (this is how it will need to face to attach to the IP67 Components Box).



- With cable gland correctly placed Solder extender wires to the end of the Waterproof cable and the other ends of the extenders to the 2.1mm Barrel Connector. (Use Heat Shrink tubes or Electrical tape to cover bare wire connections.)

- The Barrel Connector will plug in to the 1.5A USB/DC/Solar LiPo Charger Housed in the Components Box

Attaching Solar Panel Assembly to Component Box

- Make Sure the Soldered Connections on the Back of the Solar Panel are Water proofed (Sealed).

- Temporary attachment to the Box Can be done with Velcro or Hot glue dabs.

- Permanent attachment Can be done Using Clear Caulk around the entire external part of the 3-D Printed holder

(This method may make the Panel unable to be Replaced in case of Failure or damage)



6/4/2025

Assembly Continued (Part by Part)

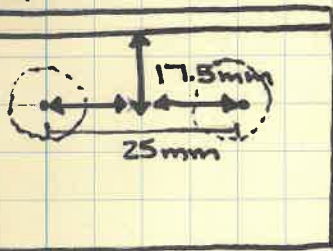


Components Box (Outside)

Attaching Cable glands is done on the left side of the Box. Cable glands allow Solar and HX711 wiring to enter box and remain Waterproof.

Front

Box Lip



2 x $\frac{1}{2}$ " holes are drilled on the left side 17.5mm from the Box Lip (down) and centered at 12.5mm each from side to side center.

- A small pilot hole for each 1st
- Then use the $\frac{1}{2}$ " drill Bit
- Fit the cable glands into place and Tighten from inside using the provided lock nut.

Attaching the Antenna is done in the Front Right corner of the Lid/Top.

- Requires a $\frac{1}{4}$ " hole to fit through.
- May Require additional Waterproofing method to the sometimes provided Rubber washer.



1 to 1.5cm
From Edge of Lid

- Drill a small pilot hole 1st
- Then use $\frac{1}{4}$ " bit
- Attach with provided washer.

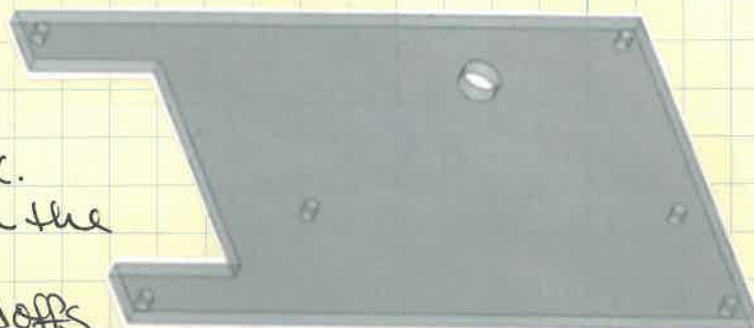
Assembly Continued (Part by Part)

6/5/2025

Components Box (inside)

- Battery Holder attaches to the Bottom of the Box Using the Provided screws.
- Attach $\frac{1}{4}$ " Standoffs to the holes at the 4 corners

The Center divider has a large cutout to accommodate the cable glands and entering cables entering on the left side of the Box.

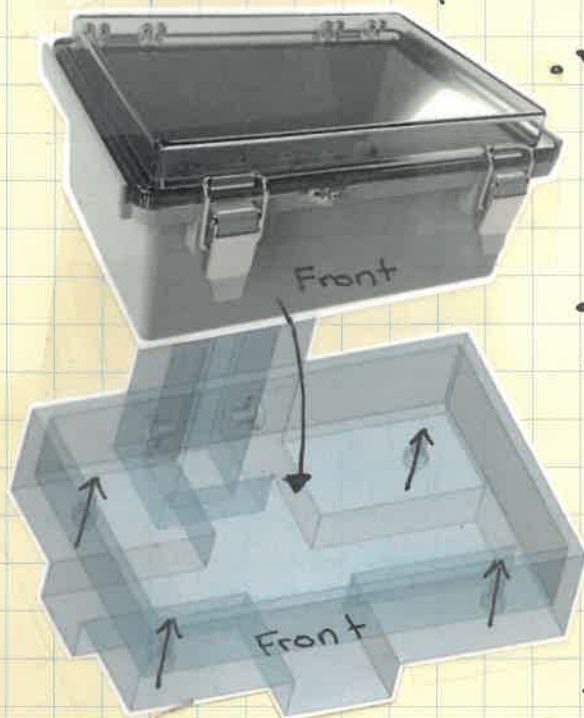


- Feed the Battery Plug through the large hole in the divider.
- Attach the divider to the Standoffs Below w/ matching Hardware in the provided holes.
- The two Additional holes are for another set of $\frac{1}{4}$ " standoffs for the Circuit Board. Attach with matching Hardware.
 - Place the circuit Board with the MKR1310 to the Front and the 1.5A LipoCharger to the left side.
- The Incoming wiring can now be attached in the manner previously provided.
- The antenna attaches directly to the MKR1310

→ Color Picture of complete Box

6/5/2025

Assembly Continued (Part by Part)

Components Box to Mount to Aluminum Rod

- Place the Components Box in the Mount and attach using 4x M4-.07x16 Stainless Screws from the Bottom side of the mount

EVERBILT

750 387

MACHINE SCREWS

M4-0.7 mm x 16 mm

PAN HEAD PHILLIPS

STAINLESS

2 PK

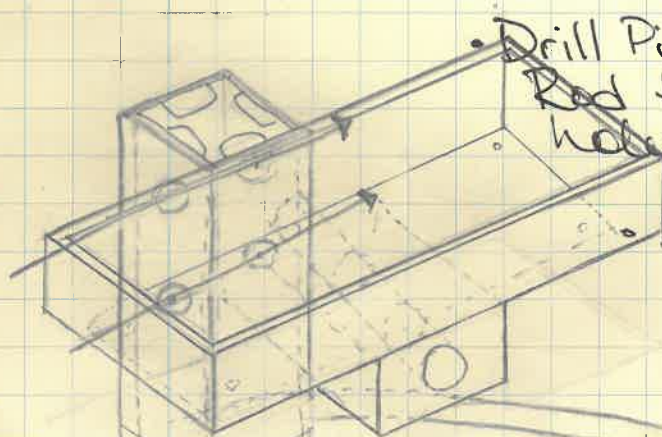


8 87480 42868 1

MADE IN TAIWAN

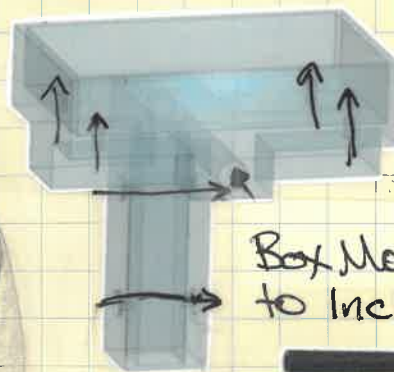
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- Slide the Mount onto the Aluminum Rod with the Box Sitting over the Floating Pipe
- With the Mount flush to the top of the Aluminum Rod Mark the Mounting hole Positions on the Aluminum Rod.



- Drill Pilot holes through the the aluminum Rod then a 1/4" drill bit to Size the holes.

- Attach with 2x 10-24x1 1/2" Stainless Steel screws and 2 Stainless Lock nuts to Tighten.



Box Mount Adapted to Include Calibration



Tool (will be stored inside Box) for doing Weighted Tank Calibration Part of Code.

Code to Create a new Data file to Save to SD Card each time the Program is ReUploaded

6/10/25

Write_Data_Into_New_SD_Card_File_Each_Upload \$

```
#include <SD.h>
#include <SPI.h>

const int CS = 11;
const char Filename[] = "H2Float";
int num = 0;
File FloatFile;

int red = 0;
int green = 1;
int blue = 4;

void setup ()
{
  pinMode(CS, OUTPUT);
  SD.begin(CS);
  delay(1000);
  if(!SD.begin(CS))
  {
    while(1);
    Failed();
  }

  bool fileExists = SD.exists("H2Float");
  while(SD.exists(Filename + String(num) + ".txt"))
  {
    num++;
  }

  void loop ()
  {
    //Saving Data to SD Card
    FloatFile = SD.open(Filename + String(num)+".txt", FILE_WRITE);
    FloatFile.println(6.7);
    FloatFile.close();
    delay(3000);
  }

  void Failed()
  {
    analogWrite(red, 255);
    analogWrite(green, 0);
    analogWrite(blue, 0);
  }
}
```

// Calls SD Card Library
// Calls SPI Library
// Chip Select at pin #11
// Create Filename
// File Number starts at zero
// Red LED at pin #0
// Green LED at pin #1
// Blue LED at pin #4
// Set Chip Select to Output
// Initialize SD Card
// Initialization failure to SD Card will hold in this loop
// Solid Red Light indicates initialization failure
// Check if the file name already exists
// If file name already exists name new
// file with the same name + 1. ".txt"
// makes the file read as a text file.
// Open File
// Insert Data
// Close file
// SD Card Fails solid RED LED
// Red out of tricolor LED

Code For Making Different Colors w/ Tri Color LED

3_Color_LED \$

```
int red = 0;
int blue = 4;
int green = 1;

void setup()
{
  pinMode(0, OUTPUT);
  pinMode(1, OUTPUT);
  pinMode(4, OUTPUT);
  Blink At Start();
}

void loop()
{
  //Red
  analogWrite(red, 255);
  analogWrite(blue, 0);
  analogWrite(green, 0);
  delay(1000);
}
```

```
//Orange
analogWrite(red, 255);
analogWrite(blue, 0);
analogWrite(green, 25);
delay(1000);
```

```
//Yellow
analogWrite(red, 160);
analogWrite(blue, 0);
analogWrite(green, 100);
delay(1000);
```

```
//Green
analogWrite(red, 0);
analogWrite(blue, 0);
analogWrite(green, 255);
delay(1000);
```

```
//Aqua
analogWrite(red, 0);
analogWrite(blue, 125);
analogWrite(green, 125);
delay(1000);
```

```
//Blue
analogWrite(red, 0);
analogWrite(blue, 255);
analogWrite(green, 0);
delay(1000);
```

```
//Purple
analogWrite(red, 120);
analogWrite(blue, 130);
analogWrite(green, 0);
delay(1000);
}
```

```
void Blink At Start()
{
  for(int j = 0; j < 10; j++)
  {
    //Blue
    analogWrite(red, 0);
    analogWrite(blue, 255);
    analogWrite(green, 0);
    delay(100);
    analogWrite(red, 0);
    analogWrite(blue, 0);
    analogWrite(green, 0);
    delay(100);
  }
}
```


6/11/2025

```

Complete_H2Float_Device_Program $
#include <Q2HX711.h> // Calls HX711 Library
#include <SD.h> // Calls SD Card Library
#include <SPI.h> // Calls SPI Library
#include <LoRa.h> // Calls LoRa Library
#include <RTCZero.h> // Calls Real Time Clock Library
#include <ArduinoLowPower.h> // Calls Low Power Library

Q2HX711 HX711(2,3); // Identifies HX711 pins (2 Black, 3 White)
long Tare;
long WeightedTare;

RTCZero rtc; // Creates an instance (name) for the RTC stuff
const byte seconds = 00; // These 3 set the start time to 00:00:00
const byte minutes = 00;
const byte hours = 00;

const int CS = 7; // Chip Select at pin #7
const char Filename[] = "H2Float"; // File name for SD Card
int num = 0; // File Number starts at zero
File FloatFile; // Create an instance for the data file

float Battery; // Create a variable for the battery voltage

int red = 0; // Red LED at pin #0
int green = 1; // Green LED at pin #1
int blue = 4; // Blue LED at pin #4

void setup()
{
  pinMode(red, OUTPUT);
  pinMode(green, OUTPUT);
  pinMode(blue, OUTPUT);
  pinMode(CS, OUTPUT);

  BlinkAtStart(); // Calls Blink at Start Program (void)
  delay(3000);
  Calibration(); // Calls Calibration Program (void)
  rtc.begin(); // Initialize RCT 24 hour format
  rtc.setTime(hours, minutes, seconds); // Sets starting clock time to 00:00:00
  if(!LoRa.begin(915E6)) // Failure, LoRa will hold in this loop 915 MHz
  {
    Failed(); // Solid Red Light indicates failure
    while(1);
  }
  SD.begin(CS); // Initialize SD Card
  delay(3000);
  if(!SD.begin(CS)) // Failure to SD Card will hold in this loop
  {
    FailedSD(); // Solid Red Light indicates failure
    while(1);
  }
  bool fileExists = SD.exists("H2Float"); // Check if the file name exists...
  while(SD.exists(Filename + String(num) + ".txt")) // If so name new file +1. ".txt" = text file.
  {
    num++; // Adds 1 to previous file name
  }
}

void loop()
{
  long Reading = HX711.read(); // Each time HX711 takes load cell reading
  float Mass = 0.75*(Reading-Tare)/(WeightedTare-Tare); // Calculate Mass
  float Force = Mass*9.8; // Calculate Force
  float Depth = (Force/18.5)*100.0; // Creates Water Depth Variable (cm)
  Battery = measureEV(); // Creates battery voltage variable (V)
  int hours = rtc.getHours(); // Retrieve RTC Hours data for use
  int minutes = rtc.getMinutes(); // Retrieve RTC Minutes data for use
  int seconds = rtc.getSeconds(); // Retrieve RTC Seconds data for use
  String TimeString = ("Time: " + String(hours) + ":" + String(minutes) + ":" + String(seconds)); // Creates RTC variable (hh:mm:ss)
  // Saving Data to SD Card
  FloatFile = SD.open(Filename + String(num) + ".txt", FILE_WRITE); // Declare Create Open new file
  FloatFile.println(TimeString+"\t"+Depth+"\t"+Battery Voltage: "+(Battery)+"V"); // Print Data to the file
  FloatFile.close(); // Close the file
  //Start LoRa and Make Packets
  LoRa.begin(915E6); // Start LoRa each loop
  LoRa.beginPacket(); // Creates packet to send
  LoRa.print('H'); // Header to specify H2Float data
  LoRa.print('D'); // Header to identify Depth specific data
  LoRa.print(Depth); // Data to send in packet
  LoRa.endPacket(); // Closes packet
  delay(1000);
  LoRa.beginPacket(); // Creates packet to send
  LoRa.print('B'); // Header to specify H2Float data
  LoRa.print('B'); // Header to identify Battery specific data
  LoRa.print(Battery); // Data to send in packet
  LoRa.endPacket(); // Closes packet
  delay(1000);
  // Shut down LoRa
  digitalWrite(LORA_RESET, LOW); // Take it low to conserve Power
  digitalWrite(LORA_BOOT0, LOW);
  pinMode(LORA_RESET, INPUT);
  pinMode(LORA_DEFAULT_SS_PIN, INPUT);
  pinMode(LORA_IRQ_DUMB, INPUT);
  pinMode(LORA_BOOT0, INPUT);
  // Sleep the MKR1310
  LowPower.sleep(60000); // Sleep MKR1310, Can't reload, 2xclick button
  BlinkAtWakeLoRa(); // Calls Blink at Wake LoRa Program (void)
}

float measureEV() // Creates battery voltage variable
{

```


6/11/2018

```

float Volts = 2.0*3.3*analogRead(A6)/1023.0;
return Volts;
}

void BlinkAtStart()
{
  for(int j = 0; j < 5; j++)
  {
    analogWrite(red, 0);
    analogWrite(blue, 255);
    analogWrite(green, 0);
    delay(100);
    analogWrite(red, 0);
    analogWrite(blue, 0);
    analogWrite(green, 0);
    delay(100);
  }
}

void Calibration()
{
  analogWrite(red, 255);
  analogWrite(green, 25);
  analogWrite(blue, 0);
  long Sum = 0;
  for(int ii = 0; ii < 100; ii++)
  {
    long Reading = HX711.read();
    Sum = Sum + Reading;
  }
  long Average = Sum/100;
  Tare = Average;
  analogWrite(red, 0);
  analogWrite(green, 0);
  analogWrite(blue, 0);
  delay(1000);
  BlinkForWeight();
  analogWrite(red, 120);
  analogWrite(green, 0);
  analogWrite(blue, 130);
  long WeightedSum = 0;
  for(int ii = 0; ii < 100; ii++)
  {
    long Reading = HX711.read();
    WeightedSum = WeightedSum + Reading;
  }
  long WeightedAverage = WeightedSum/100;
  WeightedTare = WeightedAverage;
  analogWrite(red, 0);
  analogWrite(green, 0);
  analogWrite(blue, 0);
}

void BlinkForWeight()
{
  for(int j = 0; j < 20; j++)
  {
    analogWrite(red, 120);
    analogWrite(blue, 130);
    analogWrite(green, 0);
    delay(500);
    analogWrite(red, 0);
    analogWrite(blue, 0);
    analogWrite(green, 0);
    delay(500);
  }
}

void Failed()
{
  analogWrite(red, 255);
  analogWrite(green, 0);
  analogWrite(blue, 0);
}

void FailedSD()
{
  for(int j = 0; j < 3; j++)
  {
    analogWrite(red, 255);
    analogWrite(green, 0);
    analogWrite(blue, 0);
    delay(100);
    analogWrite(red, 0);
    analogWrite(green, 0);
    analogWrite(blue, 0);
    delay(100);
  }
}

void BlinkAtWakeLoRa()
{
  for(int j = 0; j < 3; j++)
  {
    analogWrite(red, 0);
    analogWrite(green, 255);
    analogWrite(blue, 0);
    delay(100);
    analogWrite(red, 0);
    analogWrite(green, 0);
    analogWrite(blue, 0);
    delay(100);
  }
}

```

// Calculates current battery voltage

// Blue LED Blink Function 5 times on start up
// Indicates Uploaded or Reuploaded program
// Blinks Blue LED 5 times

// System Calibration Function
// Solid Orange LED for calibrating tare

// Takes 100 readings of the tare

// Averages the tare readings
// Creates Tare variable from the average
// Turns off Orange LED

// Calls 30 second Purple LED blink Function
// Solid Purple LED for calibrating weighted tare

// Takes 100 readings of the weighted tare

// Averages the weighted tare readings
// Creates WeightedTare variable from the average
// Turns off Purple LED

// Purple LED Blink Function
// Indicates weight needs to be added to the tare
// Blink Purple LED 20 times (20 Seconds)

// Solid Red LED Function
// Indicates failure to launch LoRa or SD Card

// Solid Red LED Function
// Indicates failure to launch LoRa or SD Card

// Blink Green LED Function
// Indicates LoRa exits lowpower mode
// Blink Green LED 3 times

6/12/2025

Program For Receiving LoRa Side of H₂Float

Complete_LoRa_Receiving

```

#include <SD.h>
#include <SPI.h>
#include <LoRa.h>
#include <RTCZero.h>
RTCZero rtc;
const byte Seconds = 00;
const byte Minutes = 12;
const byte Hours = 10;
const int CS = 11;
float Depth;
float Battery;
const char Filename[] = "H2Float";
int num = 0;
File FloatFile;

// Calls SD Card Library
// Calls SPI Library
// Calls LoRa Library
// Calls RTC Library
// Create a name (instance) for the rtc stuff
// Set the current second
// Set The current minute
// Set the current hour
// Chip Select at pin #11

// Create Filename
// File Number starts at zero

void setup ()
{
    pinMode(CS, OUTPUT);
    rtc.begin();
    rtc.setTime(Hours, Minutes, Seconds);
    LoRa.begin(915E6);
    delay(1000);
    if(!LoRa.begin(915E6))
    {
        while(1);
    }
    SD.begin(CS);
    delay(1000);
    if(!SD.begin(CS))
    {
        while(1);
    }
    bool fileExists = SD.exists("H2Float");
    while(SD.exists(Filename + String(num) + ".txt"))
    {
        num++;
    }
    // Set Chip Select to Output
    // Initialize 24 hour format
    // Sets the clock start time
    // 915 MHz
    // If LoRa fails program will stay in this loop
    // Initialize SD Card
    // Initialization failure to SD Card will hold in this loop
    // If SD Card fails will stay in this loop
    // Check if the file name already exists
    // If file name already exists name new
    // file with the same name + 1. ".txt"
    // makes the file read as a text file.

void loop ()
{
    int h = rtc.getHours();
    int m = rtc.getMinutes();
    int s = rtc.getSeconds();

    // Read Hours off the clock
    // Read Minutes off the clock
    // Read Seconds off the clock

    float packetSize = LoRa.parsePacket();
    if(packetSize > 0)
    {
        // Get the packet
        // If there is something in the packet
        char dataHeader = LoRa.read();
        if(dataHeader == 'H')
        {
            // Read the packet header
            char dataType = LoRa.read();
            if(dataType == 'D')
            {
                // Read the data type
                // If data type is D
                Depth = LoRa.parseFloat();
                // Print out the depth measurement
            }
            else if(dataType == 'B')
            {
                //If the data type is B
                Battery = LoRa.parseFloat();
                // Print out the battery voltage
            }
        }
        //Saving Data to SD Card
        FloatFile = SD.open(Filename + String(num) + ".txt", FILE_WRITE); // Open File
        if(h < 10)
        {
            FloatFile.print("0");
        }
        FloatFile.print(h);
        FloatFile.print(":");
        if(m < 10)
        {
            FloatFile.print("0");
        }
        FloatFile.print(m);
        FloatFile.print(":");
        if(s < 10)
        {
            FloatFile.print("0");
        }
        FloatFile.print(s);
        FloatFile.print("\t\t\t");
        FloatFile.print(Depth);
        //Put stuff in the file
        FloatFile.print("cm");
        FloatFile.print("\t\t\t");
        FloatFile.print(Battery);
        FloatFile.println("\n");
        FloatFile.close();
        //Close file
    }
}

```


Directions to Run H₂Float With Out Serial Monitor

Instructions for H₂Float

*Run line from floating pipe through wheel on device

3 x Blue flashes

- Upload from computer.
- Reload use MKR1310 button push.

Solid Orange

- Running calibration, no weight.

20 x Purple flashes (20 seconds)

- Time to hang 750g weight.

Solid Purple

- Running calibration, with 750g weight.

* Remove weight when Purple Light goes off.

- Calibration is complete.

Solid Red

- LoRa Failed.
- Restart program.

3 x Red flashes

- SD Card Failed.
- Check that SD Card is inserted properly
- Check SD Card for space.

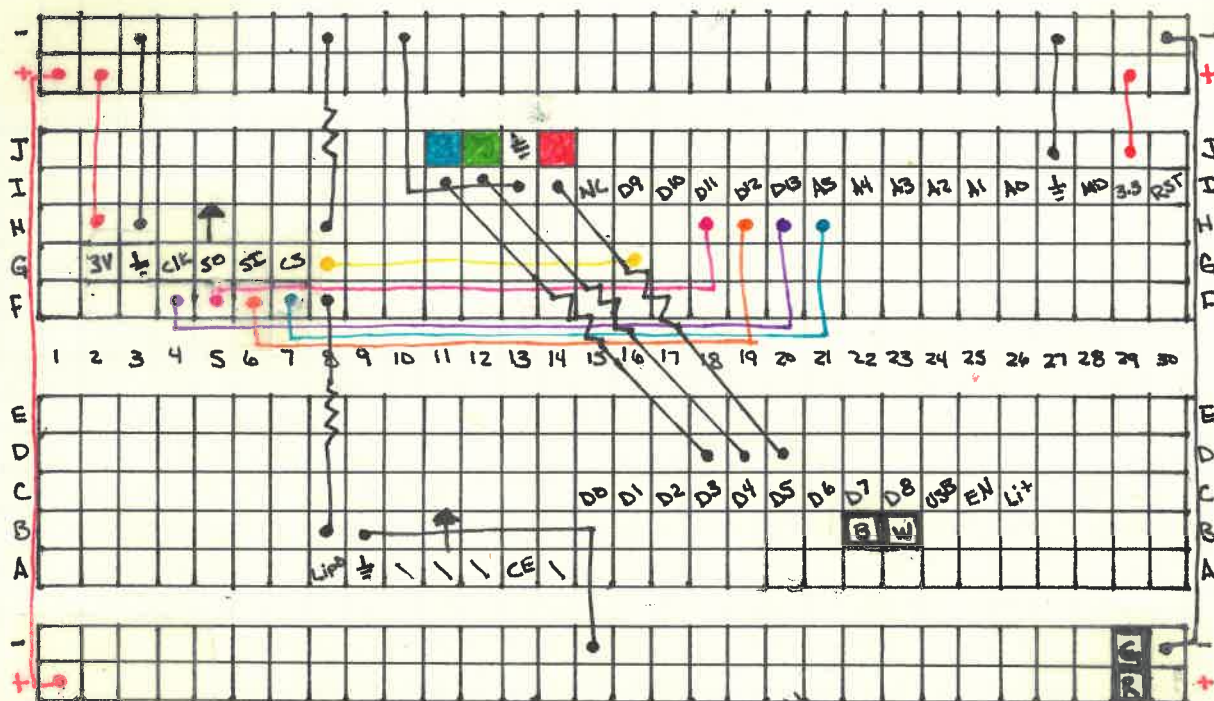
*Every Upload / Reload creates New File

- Can only have 10 files (0 to 9)
- Restart program.

3 x Green flashes (approx. every 15 minutes)

- MKR1310 board Woke from sleep mode.
- H₂Float is taking a measurement.
- Program is running.

7/10/2025 Circuit Board Design and Component Connections Particle Boron Board Specific



Board Components

Particle Boron MicroController
1.5A USB/DC/Solar LipoCharger
MicroSD Card Reader Breakout
Tri-Color LED (4-Pins)

— Wire
— Resistor

Wire Standoff

Hx-711 Amplifier Wires

B Black → DOUT → Pin#7
W White → SCLK → Pin#8
G Green → Ground
R Red → Power 3V

SD Card Breakout

CLK → Pin# D13
SD MISO → Pin# D11
SI MOSI → Pin# D12
CS ChipSelect → Pin# A5

Tri-Color LED

1. Red → Pin# D5
2. Ground → Wire to $\frac{1}{2}$
3. Green → Pin# D4
4. Blue → Pin# D3



Connected By 220Ω Resistors

Solar Charger

Lipo → 200kΩ Voltage
Divider to Pin#
D9



L9.9mm D 3.6mm H

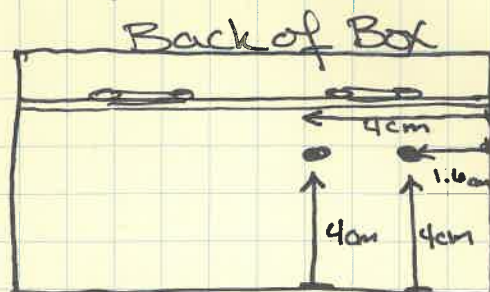
D 15.24

Din 411

6.3 2.5 10.16 H

Din 207

Large Components Box Cable Glands Positions



$\frac{1}{2}$ " hole for Cable glands
Start w/ small pilot hole
and Step Up to Larger
Sizes to keep Box from
Cracking.

→ These will eventually need to be Caulked
to ensure Water Proofing → Humidity can
still get through.